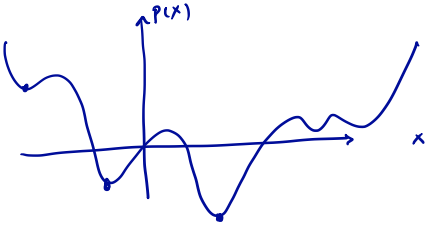
SDPs and Sums of Squares: an example

<u>Problem:</u> $p(x) = p_0 + p_1 x^1 + p_2 x^2 + \cdots + p_d x^d$

min: $p(x)$

st: $x \in \mathbb{R}^1$



$d = 2k$

$p_d > 0$

Non-convex in x.

Nonnegative Polynomials

FACT:

A poly $q(x)$ is non-negative on all of $\mathbb{R}$, <u>iff</u> $q(x)$ = sum of squares of other poly.

$$q(x) \geq 0 \quad \forall x \in \mathbb{R} \iff \exists\, h^j(x) \text{ s.t. } q(x) = \sum (h^j(x))^2.$$

(" $\Leftarrow$ " obvious.)

Checking SoS via SDP feasibility — 1/2

$$q(x) = q_0 + q_1 x^1 + \cdots + q_d x^d, \qquad d = 2k$$

Check: $\underbrace{q(x)}_{\deg\, d=2k} \overset{?}{=} \sum \underbrace{(h^j(x))^2}_{\deg \le k}$.

$$\underline{X} = \begin{pmatrix} 1 \\ x \\ x^2 \\ \vdots \\ x^k \end{pmatrix}$$

$k \times 1$

$$h^j(x) = h^j_0 + h^j_1 x + h^j_2 x^2 + \cdots + h^j_k x^k.$$

$$= \langle h^j, X \rangle \qquad \text{where} \quad h^j = \begin{pmatrix} h^j_0 \\ h^j_1 \\ \vdots \\ h^j_k \end{pmatrix}$$

$$= \langle X, h^j \rangle$$

$$\sum_j (h^j(x))^2 = \sum_j (X^T h^j)^2 = \sum_j X^T h^j h^{j\,T} X = \underset{1\times k}{X^T} \Big[\overbrace{\sum_j \underset{k\times 1}{h^j} \underset{1\times k}{h^{j\,T}}}^{k\times k} \Big] X = X^T \underset{\text{PSD.}}{Q} X$$

Checking SoS via SDP feasibility — 2/2

$$\sum (h^j(x))^2 = X^T \left(\sum h^j h^{jT}\right) X = X^T Q X$$

Using eigenvalue decomposition: if $Q \succeq 0$ then

$$X^T Q X = \text{sum of squares.}$$

Mistake

X is $(k+1)\times 1$

$$\underbrace{1\times k \quad k\times k \quad k\times 1}_{1\times 1} \qquad (1\times(k+1))((k+1)\times 1)$$

Hence: $q(x)$ is a sum of squares iff $\boxed{q(x) = X^T Q X}$ for $Q \succeq 0$.

$$q(x) = q_0 + q_1 x^1 + \dots + q_d x^d, \quad d = 2k.$$

$$X^T Q X = (1 \ x \ \cdots \ x^k) \begin{pmatrix} Q_{00} & Q_{01} & Q_{02} & \cdots & Q_{0k} \\ Q_{10} & Q_{11} & Q_{12} & \cdots & \\ Q_{20} & Q_{21} & Q_{22} & \cdots & \\ \vdots & & & \ddots & \end{pmatrix} \begin{pmatrix} 1 \\ x \\ \vdots \\ x^k \end{pmatrix}$$

$$\boxed{\begin{aligned} Q_{00} &= q_0 \\ Q_{10} + Q_{01} &= q_1 \\ Q_{20} + Q_{11} + Q_{02} &= q_2 \\ &\vdots \end{aligned} \qquad Q \succeq 0}$$

$q(x)$ is a sum of squares iff ← this SDP is feasible

Minimizing a 1-d Polynomial via SDP

$$\begin{aligned} \min:\ & p(x) \\ \text{st}:\ & x \in \mathbb{R} \end{aligned} \quad \Leftrightarrow \quad \begin{aligned} \max:\ & \gamma \\ \text{st}:\ & p(x) - \gamma \geq 0 \quad \forall x \in \mathbb{R} \end{aligned}$$

$$\Leftrightarrow \quad \begin{aligned} \max:\ & \gamma \\ \text{st}:\ & p(x) - \gamma = \sum (h^j(x))^2 \end{aligned}$$

$$\Leftrightarrow \quad \begin{aligned} \max:\ & \gamma \\ \text{st}:\ & \underline{p(x) - \gamma} = \underline{x^T Q x} \\ & Q \succeq 0 \end{aligned} \quad \Leftrightarrow \quad \boxed{\begin{aligned} \max:\ & \gamma \\ \text{st}:\ & Q_{00} = P_0 - \gamma \\ & Q_{10} + Q_{01} = P_1 \\ & Q_{20} + Q_{11} + Q_{02} = P_2 \\ & \vdots \\ & Q \succeq 0 \end{aligned}}$$

SDP!

Variables are γ & Q.